

Free Value Or Else

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Abstract

A set of free functions implementing the pattern of optional and expected `value_or` as free functions that produce the correct common reference type that was unfixable for `std::optional::value_or`.

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Changes Since Last Version

— R0 Initial revision.

1 Comparison table

1.1 Raw pointer with a manual null check

<MINTED> <MINTED>

1.2 The `value_or` truncation pitfall

<MINTED> <MINTED>

1.3 Reference alternative without copies

<MINTED>

<MINTED>

2 Motivation

3 Design

3.1 The nullable concept

3.2 Return type computation

3.3 reference_or

3.4 or_invoke and laziness

3.5 Why free functions

3.6 expected interaction

3.7 What about yield_if

4 Principles for Reification of Design

5 Implementation Experience

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6 Proposal

7 Wording

8 Impact on the standard

9 Acknowledgments

10 Document history

— R0 2026-06-12 — Initial draft. The free-function follow-on promised in [P2988R12], continuing [P1255R13].

References

[P1255R13] Steve Downey. P1255R13: A view of 0 or 1 elements: views::nullable and a concept to constrain maybes. <https://wg21.link/p1255r13>, 5 2024.

[P2988R12] Steve Downey and Peter Sommerlad. P2988R12: std::optional<t&>. <https://wg21.link/p2988r12>, 4 2025.